Virchows Archiv (2022) 481:443–451
https://doi.org/10.1007/s00428-022-03369-6

ORIGINAL ARTICLE

Deglycosylation of pathological specimens alters performance of diagnostic PDL1 antibodies

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Received: 22 April 2022 / Revised: 17 June 2022 / Accepted: 22 June 2022 / Published online: 2 July 2022
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Abstract

Immunohistochemical (IHC) predictive quantitation of PDL1 expression is obligatory in many cancer entities with improved response to immune checkpoint inhibition in PDL1-positive subgroups. With recent demonstration of increased positivity rates after enzymatic deglycosylation in breast cancer specimens, a comparative analysis with two different antibodies and extended controls was performed in a cohort of head and neck squamous cell cancer samples (HNSCC).

Formalin-fixed paraffin-embedded tissue from HNSCC specimens was used for initial on-slide method optimization based on the PNGase F assay. SDS-PAGE and immunoblotting with the PDL1 antibody 28–8 was performed to evaluate deglycosylation efficiency. A tissue micro array of n = 527 tissue cores of 181 patients with HNSCC was used to determine the effects of deglycosylation on staining pattern and intensity with PDL1 antibodies 28–8 and E1L3N.

Successful on-slide deglycosylation with PNGase F was confirmed by immunoblot but varied across replicates. Using E1L3N (intracellular binding domain, most probably not glycosylated), mean signal intensity as well as the fraction of PDL1 positive cells was increased by deglycosylation. Opposite effects were observed with 28–8 (extracellular binding domain, glycosylated).

Deglycosylation reduces diagnostic performance of the PDL1 antibody 28–8. In contrast, effects for E1L3N are complex and probably involve reduction of off-target binding leading to specifically improved signal intensity. However, enzymatic deglycosylation adds further variance to IHC.

Keywords PDL1 diagnostics · Deglycosylation · HNSCC · Diagnostic performance

Introduction

The introduction of immune checkpoint inhibitors (ICI) changed therapeutic regimens in several cancer entities. Especially in melanoma treatment, the blockade of the programmed cell death protein 1 (PD1) on lymphocytes or the programmed death-ligand 1 (PDL1) leads to significantly improved survival [1]. However, the efficacy of ICI therapy differs across cancer entities.

In head and neck squamous cell carcinoma (HNSCC), median overall survival in the pre-ICI era in patients with recurrent or metastatic disease was 10 to 13 months [2, 3]. Based on its success in melanoma therapy, expectations were quite high for ICI treatment in HNSCC but improvement of overall survival was only moderate [4–6]) and the effect on actual long-term survival remains unclear [7]. Currently, the PD1 inhibitors Nivolumab and Pembrolizumab are the only approved drugs in HNSCC. PDL1 expression quantified by immunohistochemistry (IHC) and the combined

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positive score (CPS) serves as a limited predictive marker for Pembrolizumab, while it appears not to be predictive for Nivolumab [4, 7].

While different diagnostic PDL1 antibodies have been used [8], the lack of reliable predictive PDL1 quantitation in HNSCC in contrast to other malignancies cannot be explained by this factor alone. Recently, differences in glycosylation patterns — already a known factor in cancer and drug response [9] — have been linked to altered diagnostic performance [10]. In non-small cell lung cancer and triple-negative breast cancer PDL1 glycosylation can be affected by the EGFR signaling pathway [11, 12]. With EGFR overexpression frequently observed in HNSCC as well, this could be a relevant factor for inconsistent PDL1 quantification [13]. In several other cancer entities, Lee et al. [14] in fact showed that enzymatic deglycosylation before IHC can improve predictive PDL1 diagnostic performance.

Here, we investigate and optimize the methodological approach of pre-IHC deglycosylation. Using two diagnostic PDL1 antibodies, we evaluate the deglycosylation effects across a cohort of HNSCC primaries. While we identify methodological challenges, we observe deglycosylation effects that contrast with previously reported findings.

Materials and methods

Study cohort

295 primary tumors were selected from our previously described HNSCC cohort [15]. After accounting for technical and staining artifacts, a final number of 181 primary tumors were further evaluated. Ethical approval was obtained from the Ethics Committee of the University of Luebeck (16–277). The study was conducted in accordance with the Declaration of Helsinki. The selected tumors were arranged in four tissue micro arrays (TMA) as previously described [16].

Deparaffinization and enzymatic deglycosylation

Whole-block or TMA sections on standard slides from formalin-fixed paraffin-embedded (FFPE) tissue were deparaffinized by immersion into a water bath at 60 °C for 30 min. After deparaffinization, the HybriWell™ sealing system (611,102; Grace Bio-labs, Bend, USA) was attached to the slides creating a sealed compartment which included the surface of the coated slide as floor. The protocol of the Protein Deglycosylation Mix II (P6044S; New England Biolabs, Ipswich, USA) was modified for on-slide deglycosylation: 80 µl of double-distilled water (ddH_2O) containing glycoprotein denaturing buffer 1 × was added to the chamber, the ports were sealed with Parafilm™ and the slides were incubated in a water bath at 90 °C for 10 min. After cooling to room temperature, the cover was removed, and the slides were centrifuged at 1000 G for 1 min, upright in a 50 ml tube to remove the buffer. The empty chamber was then filled with 80 µl ddH_2O containing NP-40 10% v/v, GlycoBuffer2 1 ×, HALT protease and phosphatase inhibitor cocktail 1% v/v (78,440 Thermo Scientific, Waltham, USA) as well as 4 µl PNGase F (from P6044S as well) for deglycosylated samples. The slide chamber ports were closed with adhesive tape and all slides were incubated overnight at 37 °C. After incubation, the chambers were removed and the slides washed three times in PBS.

Protein extraction

After deparaffinization and enzymatic deglycosylation as described above, the tissue was scratched from the slides into 0.2 ml PCR tubes and 50 µl extraction buffer was added (HALT protease Inhibitor 1% v/v (78,440 Thermo scientific, Waltham, USA), ß-Mercaptoethanol 5% v/v, sodium dodecyl sulfate 2% w/v, Tris-base 200 mM, EDTA 1 mM, pH 7.2). The samples were then incubated on ice for 5 min, vortexed, placed in a thermocycler, incubated at 4 °C for 5 min, 90 °C for 90 min and then 4 cycles of 99 °C for 5 min followed by 60 °C for 10 min. Samples were vortexed again and undissolved parts removed by centrifugation at 10,000 G for 15 min. Supernatants were stored at – 20 °C for a maximum of 1 week.

Immunoblot analysis

Immunoblotting was performed with a 10% MOPS-SDS Bis–tris Gel (NW00105BOX, ThermoFisher Scientific, Waltham, USA) using the Xcell SureLock™ system (EI0001 ThermoFisher Scientific, Waltham, USA) at 200 V (maximum 250 mA) for 35 min with 10 µl protein extract per lane (corresponding to 20 µg per lane). Transfer was at 30 V (maximum 250 mA) for 60 min on nitrocellulose membrane (LC2001 ThermoFisher Scientific, Waltham, USA). Membranes were cut, washed twice with ddH_2O and blocked for 30 min in 5% w/v milk in TBS containing 0.5% v/v Tween-20 (TBS-T-M). Membranes were washed twice with milk-free TBS-T. Primary incubation was done overnight using the rabbit IgG PDL1 monoclonal antibody 28–8 (1:500, ab205921, Abcam, Cambridge, UK) with rabbit IgG anti beta integrin (1:1000, #4706, Cell Signaling Technology, Danvers, USA) and mouse IgG anti-HPRT (sc-376938, Santa Cruz Biotechnology, Dallas, USA) antibodies as loading controls. Membranes were again washed four times and incubated with the secondary antibodies anti-rabbit IgG (1:2500, #31,460, ThermoFisher Scientific, Waltham, USA and anti-mouse IgG (1:2500, #31,430, ibid.) for 120 min. Membranes were

then washed with TBS-T three times and with TBS once. Membranes were then reassembled and optically evaluated on a densitometric imager (Amersham Imager 600, GE Healthcare 29,083,461, Freiburg, Germany) using ECL developing agents (GE Healthcare Life Sciences Europe RPN2106, Eindhoven, Netherlands).

Immunohistochemistry and evaluation of stained slides

Immunohistochemistry (IHC) was performed as previously described [17]. Two primary PDL1 antibodies were used, 28–8 (1:100, ab205921, Abcam, Cambridge, UK) and E1L3N (1:50, 13,684, Cell Signaling Technology, Danvers, USA). The OptiView Diaminobenzidine (DAB) detection kit was implemented on a Ventana Benchmark Ultra automated staining system (Roche, Basel, Switzerland). The slides were digitalized on a Ventana iScan HT scanner (Ventana, Tuscon, USA) with a 20-fold objective. For digital evaluation and quantification, QuPath 0.2.3 was used [18]. Tumor areas were manually segmented as regions of interest of the TMA cores, maximizing tumor cell purity (minimizing immune cells, stromal cells, blood vessels, etc.). DAB intensities were automatically measured as optical density. Thresholds were set based on visually controlled standard thresholds for HE staining at 0.05 (dimensionless QuPath scores) and for immunohistochemical staining at 1.6 (idem) for the cells to be considered as PD-L1 positive cells. The analyzed images were independently reviewed afterwards by a board-certified pathologist before downstream statistical analysis.

Statistical analysis

All statistical analyses and data visualizations were performed in Python 2.7.17 using the packages numpy 1.16.1, scipy 1.2.2, matplotlib 2.2.4, seaborn 0.9.1, and pandas 0.24.2. Data is reported as median with interquartile range, significance was tested for using the Mann–Whitney *U* test. Statistical testing was used and is reported only for a priori defined comparisons. Strict conservative Bonferroni adjustment [19] of the standard significance level 0.05 leads to 0.016 for each a priori defined parameter [20].

Results

On-slide deglycosylation

The standard protocol for deglycosylation was successfully adopted to low-volume application on-slide. To this end, a commercially available on-slide reaction chamber was used to minimize the necessary volume and to maximize the contact surface between buffer, enzyme, and tissue. To evaluate deglycosylation success, we employed IB analysis of the molecular size of PDL1 (Fig. 1), showing a size decrease by around 15 kDa as previously reported [14]. As a positive control, beta-integrin, a membrane protein with known glycosylation [21], demonstrated a similar shift. In contrast, the cytosolic protein HPRT (glycosylation, if any, 0.3 kDa [22]) did not exhibit a reduction of its size.

We varied the standard deglycosylation protocol by the enzyme manufacturer both in-tube (data not shown) and on-slide to investigate the effects of denaturation prior

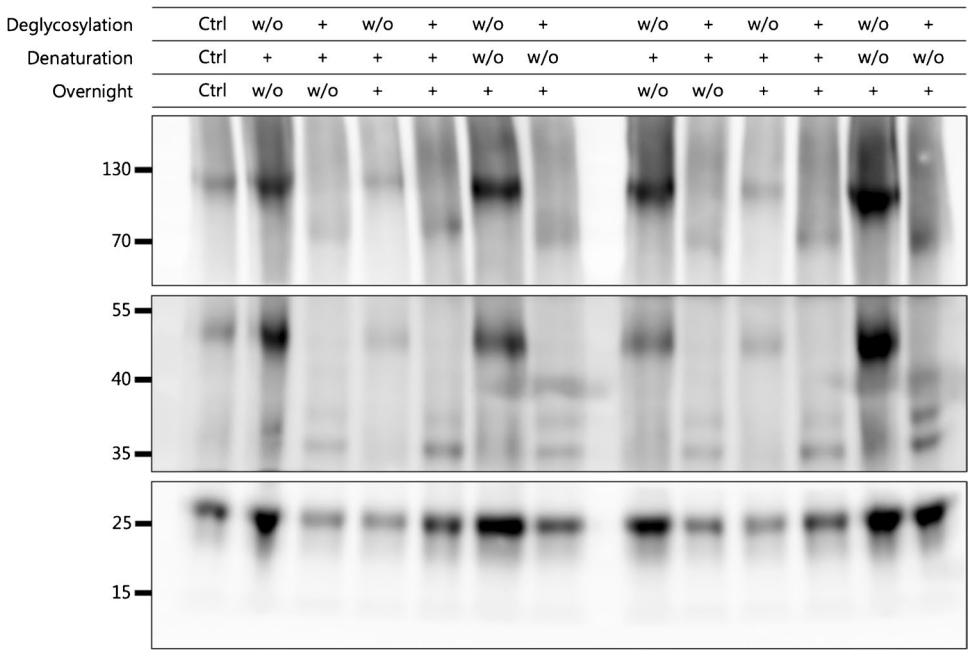


Fig. 1 *Deglycosylation protocol variants in immunoblot analysis.* The protocol variants were blotted in two technical replicates (left and right set of lanes respectively); Upper image = beta integrin; Middle image = PDL1; Lower image = HPRT; Ctrl = control sample, which was not subjected to the protocol (original protein extract); w/o = without

to deglycosylation as well as prolonged deglycosylation overnight. The effects are exemplarily shown in Fig. 1 and demonstrate incomplete deglycosylation with a stronger medium-sized band when denaturation was omitted. There was no pronounced difference concerning the duration of enzymatic deglycosylation. To parallel the previously published assay [14], deglycosylation of our cohort was also performed overnight.

Deglycosylation effects on PDL1 immunohistochemistry.

We evaluated tumor cells of N = 527 tissue cores (samples) of 181 primary HNSCC tumors, arranged in four TMAs. In the four groups, E1L3N without (w/o) or with deglycosylation (+ DG) and 28–8, respectively, semi-automated IHC image analysis of 482, 491, 394, and 475 samples detected median n = 2232, 2544, 2686, and 2644 cells per sample (significantly different only between the w/o groups; $p = 0.01$; Fig. 3D).

Figure 2 shows exemplary IHC images, with different examples of the deglycosylation effect across different TMAs. The results across the complete cohort are visualized in Figs. 3, 4, 5. A distinct, specific increase in DAB intensities upon deglycosylation could be detected for E1L3N. In contrast, 28–8 intensities were considerably lowered. The fraction of PDL1-positive cells (Fig. 3) increased for E1L3N (median 0.0068, interquartile range [0.0018–0.0293] vs. 0.0544 [0.0103–0.3748]; $p < 0.001$) and decreased for 28–8 (0.00131 [0.00043–0.00657] vs. 0.00049 [0.00028–0.00084]; $p < 0.001$). Thereby E1L3N + DG decreased the number of PDL1-negative samples (no positive cells) from 14 to 8%, conversely 28–8 + DG showed an increase from 54 to 65% (Fig. 3A). Similar, the mean DAB intensity was relevantly increased by deglycosylation for E1L3N (0.057 IQR [0.035–0.076] vs. 0.110 [0.091–0.148]; $p < 0.001$) and decreased for 28–8 (0.049 [0.042–0.059] vs. 0.041 [0.037–0.052]; $p < 0.001$; Fig. 4A). The paired comparison also showed a relevant increase of DAB intensities for E1L3N in three out of the four TMAs, particularly in samples that exhibited weak signals w/o deglycosylation (Fig. 4B). While signal intensities correlated positively in both sets and across controls w/o deglycosylation, + DG samples showed no relevant correlation (Fig. 4C–F).

Variance, measured by log standard deviation per sample (Fig. 5) was decreased by deglycosylation for both antibodies. However, two peaks were discernable in E1L3N w/o, with the higher — resulting from a particular TMA — being reduced by deglycosylation (Fig. 5C). A similar reduction was absent in the cytoplasmic DAB intensities.

Replicates of the same tumor were equally altered by deglycosylation, as indicated by comparable coefficients of variation of the mean DAB intensities (Fig. S1).

Discussion

Method validation

We implemented a protocol optimization for standardized on-slide deglycosylation with minimal amounts of enzyme. Efficient deglycosylation was controlled for by IB analysis of the actual FFPE tissue. As predicted, successful deglycosylation decreased the size of the observed PDL1 band by about 15 kDa to around 35 kDa, well in line with the theoretical weight of the deglycosylated protein (33 kDa [23]). We included both positive and negative controls.

Lee et al. [14] used several different positive controls for their study. These were analyzed with different antibodies and only in part with the actual diagnostic antibody 28–8, on which the subsequent diagnostic analyses and conclusions were based. While their IB analysis of cell lines with deglycosylation showed a similar size shift, they did not use the actual antibody 28–8 or the actual FFPE tissue for this purpose. For semi-quantitative fluorescence microscopy of formalin-fixed cell lines, the primary antibody ab51880 was used, which has been discontinued by the manufacturer due to quality concerns [24]. PDL1 quantities were further investigated in formalin-fixed cells, not the actual FFPE tissue, by ELISA assays, which do not provide molecular size or subcellular localization as internal specificity control. Additionally, the ELISA method description indicates that PNGase F pretreated samples were selectively blocked with BSA, inducing a bias by the high abundant protein. Negative controls were not used. For the actual FFPE tissue, exemplary IHC images were shown (see below).

We used four different TMAs for our method validation approach. All four included samples from the same HNSCC cohort with uniform distribution across the different TMAs. As each TMA section was processed separately, different TMAs served as independent methodological replicates. While we observed deglycosylation effects, it has to be noted beforehand that the effect size differed between the different TMAs (Fig. 2, 3C, 4B, 5C–D). While the antibody-based, enzymatic IHC method is semiquantitative per se [25], the additional enzymatic deglycosylation step appears to add further quantitative variance. As for IHC inter-run variance in general — in particular for (semiquantitative) protein expression quantitation — this variance could only be controlled qualitatively by means of external on-slide positive and negative controls.

Choice of antibodies

We evaluated two antibodies, which are both in routine diagnostic use. As the binding sites of several diagnostic antibodies have recently been mapped by Lawson et al. [26], we chose the clones E1L3N and 28–8. The latter has been investigated by Lee et al. [14] and binds to an extracellular and

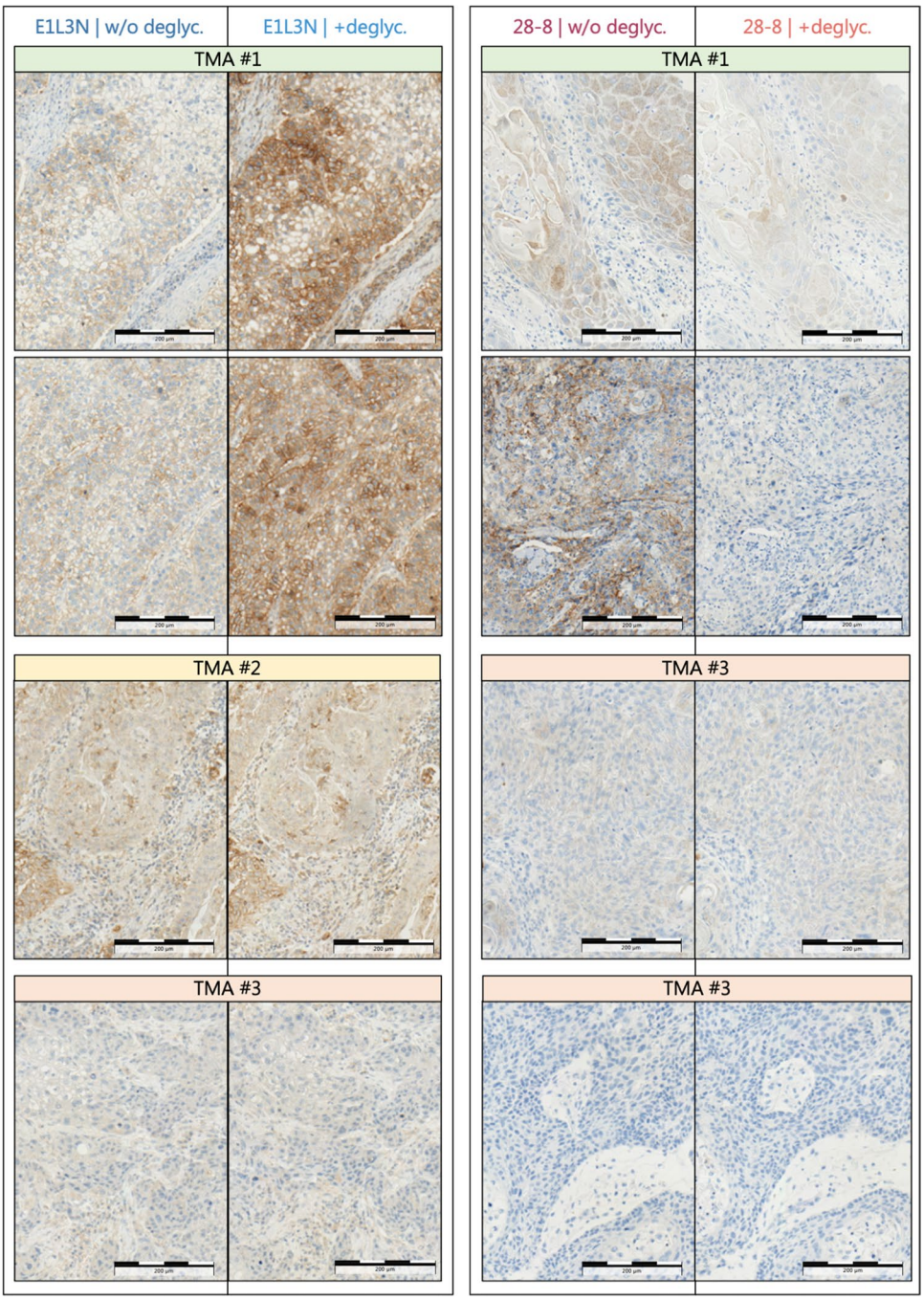


Fig. 2 *Deglycosylation effects on anti-PDL1 immunohistochemistry.* Paired exemplary microscopic images for each group and different tissue micro arrays (TMA), which represent different protocol replicates

likely glycosylated site. E1L3N is in widespread diagnostic use and binds to an intracellular region at the C-terminus between amino acids 280–290, outside of any known glycosylation site (the closest being at position 219 [23]).

Reduced performance of 28–8

We observed a decrease in DAB intensity as well as the fraction of positive tumor cells with the diagnostic PDL1 antibody 28–8. Similarly, Lawson et al. [26] showed reduced intensities of 28–8 on both IHC and IB level when N-linked glycosylation sites were targeted in PDL1 mutants. Also bearing in mind that PDL1 deglycosylation was most likely not considered at the time of the development of this antibody, it probably targets a complex epitope consisting of an entwined ravel of sugars and the protein peptide chain. Even *ex ante*, it would not have been surprising that deglycosylation rather decreases antibody binding by altering this epitope, much less so with the experimental data of Lawson's and our study.

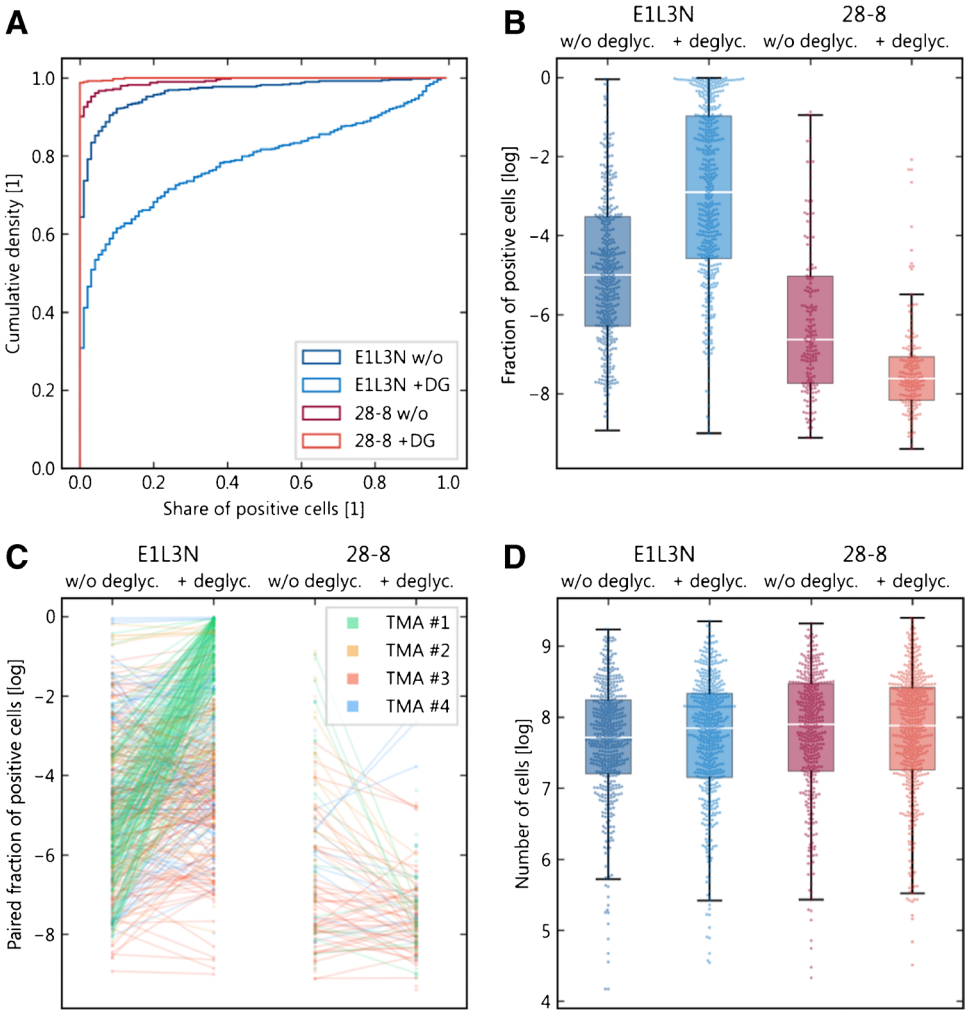


Fig. 3 *Deglycosylation effects on the fraction of PDL1-positive tumor cells.* (A) Cumulative density plots of all samples; (B) Distributions of positive fractions in samples with at least one positive cell; (C) Paired positive fractions of deglycosylated samples and their respective controls; Color-stratified by respective tissue micro array (TMA); (D) Distributions of the number of detected cells as evaluation control

In contrast to these results, Lee et al. [14] have prominently demonstrated an increase in PDL1 detection with 28–8 after deglycosylation. Close examination of their work, however, leads to a series of limitations. First, as outlined above, quantitative controls were patchy. For FFPE tissue (the material most of the study's diagnostic and clinical conclusions were based on) only exemplary IHC images were provided. Several of these (Fig. 2C, 2F, 3C, and S2D [14]) demonstrated increased signal intensity but without cellular or membrane specificity. Instead, stromal cells and cell-free background were stained too. This also renders manual evaluation of open-label slides more difficult, especially when using the H-score. To avoid this, we used two antibodies in our study, blinded the semi-automated quantitative evaluation and focused on the actual diagnostic readout (the fraction of positive cells).

Apart from these basic limitations, the diagnostic and clinical utility of sample deglycosylation for 28–8 IHC was primarily shown in correlation with survival data. Here, only *p*-values of linear regressions were compared, not the actual correlation coefficients, although Pearson's correlation was used as regression method. It must be noted that Pearson's linear regression is susceptible to outliers [27], which are intrinsically present in the skewed survival data (e.g. Figure 3E [14]). Additionally, survival was not an independent endpoint, as it had most likely been influenced by therapeutic decisions based on the initial IHC without deglycosylation (no specific explanations on time course and treatment were given). With the number of subset analyses as well as variable thresholds and without multiple testing corrections, spurious or confounded correlations cannot be excluded. On the other hand, Lee et al. investigated several different malignant entities, only partially overlapping with our study. We investigated head and neck squamous cell carcinoma primaries while Lee et al. reported some data on pulmonary squamous cell carcinoma.

Increased performance of E1L3N

We found a consistent and relevant improvement in DAB intensities for the diagnostic antibody E1L3N when samples were deglycosylated prior to IHC. The staining pattern proved to be specific on visual inspection (contrasted cell borders) and signal variance across positive cells was reduced (at least in

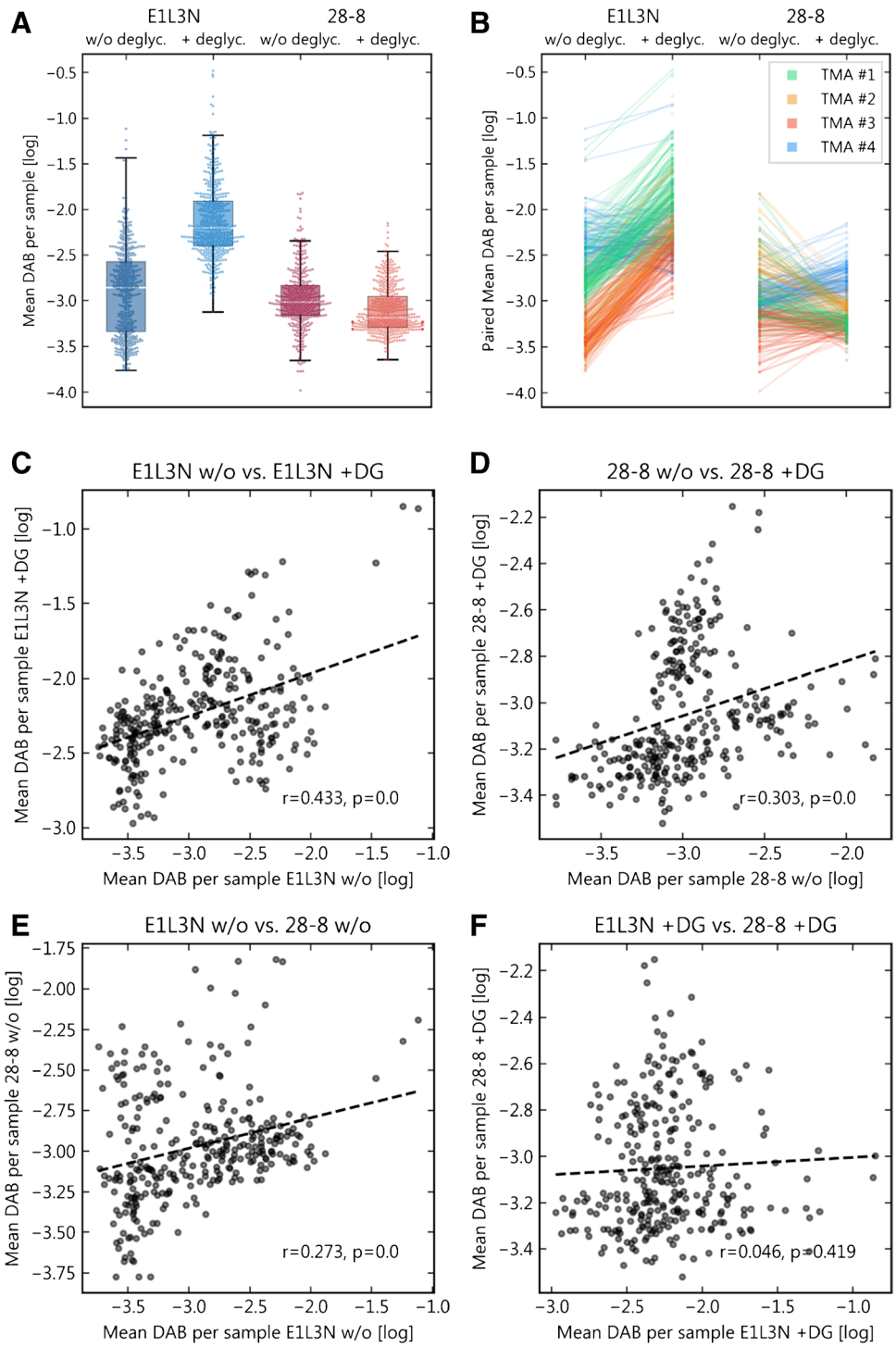


Fig. 4 *Changes of anti-PDL1 DAB intensity values.* (A) Distributions of sample-wise mean DAB values; (B) Paired deglycosylated samples and their respective control; Color-stratified by the respective tissue micro array (TMA) as in Fig. 2; (C)–(F) Correlation of mean DAB values across groups; *r* is Pearson's correlation coefficient; *p*-value from the Wald test of the linear regression

one TMA). Good correlation of E1L3N with the actual protein amount has been demonstrated by mass spectrometric (MS) targeted proteomics [28]. Interestingly, the work by Morales-Betanzos et al. also showed variant glycosylation and differences (limited by small sample size) between E1L3N IHC and MS. Higher glycosylation of the extracellular domain was found in several samples with low IHC intensity but normal or increased quantities in MS. This data indirectly supports our observation of increased PDL1 detection with deglycosylation. It remains unclear, however, whether the observed extracellular glycosylation hinders binding to the intracellular domain (unlikely in the light of the epitope mapping by Lawson et al.) or if glycosylation is generally increased in these samples, leading to non-specific and, as such, weaker off-target binding. The subsequent washing steps in IHC could then remove the loosely bound antibodies and cause reduced signal intensity.

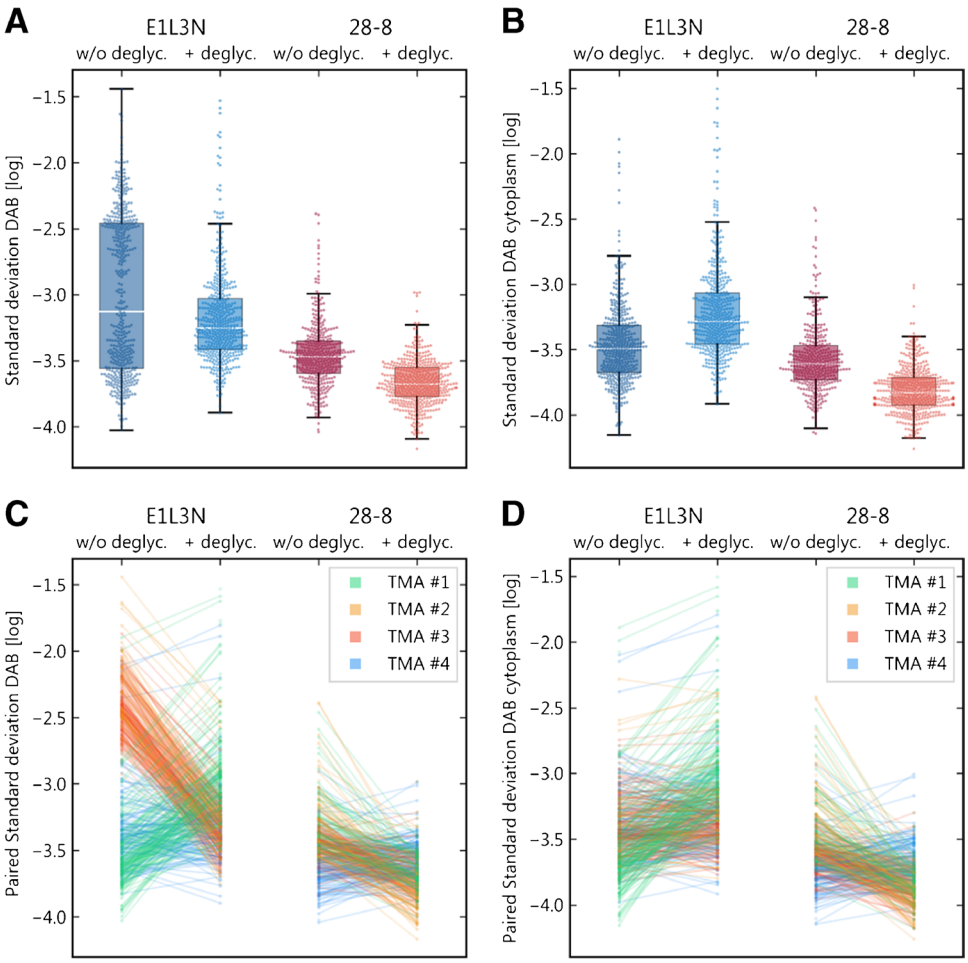


Fig. 5 *Effects on anti-PDL1 intensity variance.* (A) Distribution of the DAB standard deviation across samples; (B) In analogy to subplot A with cytoplasmic DAB intensities only; (C)–(D) Respective paired distributions; Color-stratified by respective tissue micro array (TMA)

Limitations

As outlined above, we noted relevant methodological variance for on-slide enzymatic deglycosylation. While we tried to optimize standardization, the long protocol duration with exposure to elevated temperatures will most certainly cause competing protein degradation. It cannot be excluded that subtle handling or exposure differences between actual protocol executions in different laboratories could differentially affect the balance between improved binding by deglycosylation and parallel thermal degradation.

Due to the difficulty of reliable immune cell detection in the automated image analysis, we did not include immune cells in our analysis.

Conclusions

Deglycosylation of clinical samples reduces the diagnostic performance of the PDL1 antibody 28–8, which presumably binds a glycosylated epitope, posing a hurdle to standardized diagnostics using this antibody. Conversely, the performance of E1L3N is significantly increased. Staining pattern and variance analysis hint at specific enhancement of PDL1 detection. With the binding domain of E1L3N being most likely free of glycosylation, this improvement probably originates from reduced off-target binding. While deglycosylation can thus be beneficial, the existence of this phenomenon adds further complexity to the issue of diagnostic PDL1 quantitation. Additionally, relevant differences in the deglycosylation effect size were observed across methodological replicates. Together with the known methodological challenges in IHC-based biomarker quantification, diagnostic PDL1 quantitation calls for a transition to new analytical assays of the mass spectrometric era.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00428-022-03369-6.

Acknowledgements FFD would like to thank the Else Kröner-Fresenius-Stiftung for their support.

Author contribution All named authors meet the International Committee of Medical Journal Editors (ICMJE) criteria for authorship for this article, take responsibility for the integrity of the work as a whole, and have given their approval for this version to be published.

FFD: Conceptualization, Methodology, Investigation, Formal analysis, Data Curation, Visualization, Supervision, Project administration,

Writing—Original Draft; DSD: Investigation, Data Curation, Writing—Original Draft; LK: Investigation, Data Curation, Visualization, Writing—Original Draft; SP: Writing—Review & Editing; JRI: Formal analysis, Supervision, Writing—Review & Editing; CI: Conceptualization, Writing—Original Draft.

Funding FFD received funding from the Else-Kröner-Fresenius-Stiftung (2018_A84).

Data availability Access to associated data will be granted on request to the main or senior authors.

Declarations

Conflict of interest The authors declare no competing interests.

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